Lamia Salsabil

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EDUCATION

Old Dominion University (ODU) **Norfolk, VA**
Ph.D. in Computer Science; GPA: 3.77 *Dec. 2026*
Advisor: Dr. Jian Wu

Khulna University of Engineering & Technology (KUET) **Dhaka, Bangladesh**
B.Sc. in Computer Science and Engineering *Feb. 2018*

TECHNICAL SKILLS

Languages/Libraries: Python (NumPy, Pandas, scikit-learn, PyTorch), Ruby, C, Shell, C++, R, JavaScript (jQuery, D3.js)

Web: HTML5, CSS3, Ruby on Rails (MVC), MySQL, PostgreSQL, MongoDB, Apache, Nginx, Git, JSON, XML, Agile (Scrum), Refinery CMS, SQL Server

Software: Weka, TensorFlow, Postman, Linux, Apache Spark, Amazon Textract, GROBID, Apache Hadoop

EXPERIENCE

Graduate Research Assistant **Norfolk, VA**
Old Dominion University *Jan. 2021 – Present*

- Collaborating with Virginia Tech and the Internet Archive on an IMLS-funded project to study computational reproducibility by identifying scholarly URLs linking to OADS, using a novel Ensemble OADS Classifier trained on OADS-1K (1,129 samples), achieving 90% F1-score.
  Technologies: PyTorch, TensorFlow, SciBERT, BertGCN, SPECTER, GPT-4, Claude 3.7 Sonnet.
- Designed ETD-MS v2.0, a comprehensive metadata schema for Electronic Theses and Dissertations (ETDs), addressing limitations in existing standards (Dublin Core, ETD-MS v1.1) by incorporating a richer set of ETD-specific fields and enabling support for both document-level and AI-generated content-level metadata, aligned with FAIR principles.
  Technologies: Python, BeautifulSoup, PyMuPDF, Pytesseract, MySQL, GPT-4, LDA2Vec, BERT, YOLO, T5, Pegasus.
- Built a repository of 533K+ ETDs from 114 universities using OAI-PMH.
  Technologies: Python, OAI-PMH API, MySQL.
- Designed ETDMIS, a scalable metadata enhancement system that addresses MetaEnhance's error-prone corrections by performing metadata validation using a binary classifier based on linguistic features (e.g., token count, TF-IDF), along with NER and fuzzy matching, enabling reversible updates and processing over 100K U.S. ETDs within 72 hours on a 32-core VM.
  Technologies: Python, spaCy, scikit-learn, fuzzywuzzy, PostgreSQL, Pandas, NumPy, Bash.

Software Engineer Intern **Fort Monroe, VA**
Virginia Marine Resources Commission *May 2024 – Aug. 2024*

- Redesigned FOIA website and digital tracking system to replace outdated paper-based processes.
- Conducted internal training on Visual Studio Code, TypeScript, Git, and MySQL.
- Simulated test sales for the 2025 license portal to identify and prevent sale conflicts.

Research Intern **Los Alamos, NM**
Los Alamos National Laboratory *Jun. 2023 – Aug. 2023*

– Built a framework to classify scholarly URLs from arXiv into datasets, software, media, and publishers.
– Used GROBID and regex to preprocess arXiv data and build ground truth.
– Achieved 89% F1-score using BERT, DistilBERT, and SciBERT.

Software Engineer **Dhaka, Bangladesh**
Nascenia *Jan. 2019 – Dec. 2020*

– Developed VAT processing software (Uddogi) for manufacturing companies.
Technologies: Ruby on Rails, MySQL, HTML, CSS, JavaScript.
– Built CMS for UNDP-LIUPCP using Refinery CMS.
Technologies: Ruby on Rails, MySQL, leaflet.js.
– Provisioned and deployed web servers using Nginx.

PROJECTS

- **Ensemble OADS Classifier** – An ensemble classifier that combines a Supervised Contrastive Learning model, a SciBERT-based model, and a BertGCN model to automatically classify Open-Access Data and Software (OADS) URLs in scholarly documents, distinguishing between resource types (dataset vs. software) and providers (author vs. third-party), achieving an F1-score of 90%. GitHub
Technologies: PyTorch, TensorFlow, SciBERT, BertGCN, SPECTER, Ensemble Learning.

- **OADSClassifier** – A hybrid classifier consisting of a heuristic and a supervised learning model to extract and classify URLs in scientific papers into two categories: OADS URLs (linking to open-access datasets/software) or non-OADS URLs with a best f1-score of 92%. GitHub
Technologies: Python, BERT, DistilBERT, TensorFlow, Logistic Regression, Supervised Learning.

- **Virtual Realtor** – A dialogue assistant that interacts with the user via a voice-enabled interface to find housing/apartments based on the user's requirements. GitHub
Technologies: Python, NER, REST API, Flair.

PUBLICATIONS

- Salsabil, Lamia, Jian Wu, Muntabir Hasan Choudhury, William A. Ingram, Edward A. Fox, Sarah M. Rajtmajer, and C. Lee Giles. "A study of computational reproducibility using URLs linking to open access datasets and software." In Companion Proceedings of the Web Conference 2022, pp. 784-788. 2022.

- Choudhury, Muntabir Hasan, Lamia Salsabil, Himarsha R. Jayanetti, Jian Wu, William A. Ingram, and Edward A. Fox. "Metaenhance: metadata quality improvement for electronic theses and dissertations of university libraries." In 2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL), pp. 61-65. IEEE, 2023. (Best Paper Award)

- Salsabil, Lamia, Jian Wu, William A. Ingram, and Edward Fox. "ETD-MS v2. 0: A Proposed Extended Standard for Metadata of Electronic Theses and Dissertations."

- Ingram, William A., Jian Wu, Sampanna Yashwant Kahu, Javaid Akbar Manzoor, Bipasha Banerjee, Aman Ahuja, Muntabir Hasan Choudhury, Lamia Salsabil, Winston Shields, and Edward A. Fox. "Building datasets to support information extraction and structure parsing from electronic theses and dissertations." International Journal on Digital Libraries 25, no. 2 (2024): 175-196.

- Salsabil, Lamia, Jian Wu, William A. Ingram, and Edward A. Fox. "Toward Automatically Improving Metadata Quality of Electronic Theses and Dissertations at Scale." In 2024 IEEE International Conference on Big Data (BigData), pp. 8825-8827. IEEE, 2024.

VOLUNTEER EXPERIENCE

- President, Association for Computing Machinery Women (ACM-W), ODU
- Mentor, Special Group Interested in Programming Contest (SGIPC), KUET
- Student Leader, Global Student Friendship, Norfolk